

Amrita Vishwa Vidyapeetham - School of Computing, Chennai

Topic: Tree 1 (10 Challenge Programs) | Roll No: CH.SC.U4CSE25262

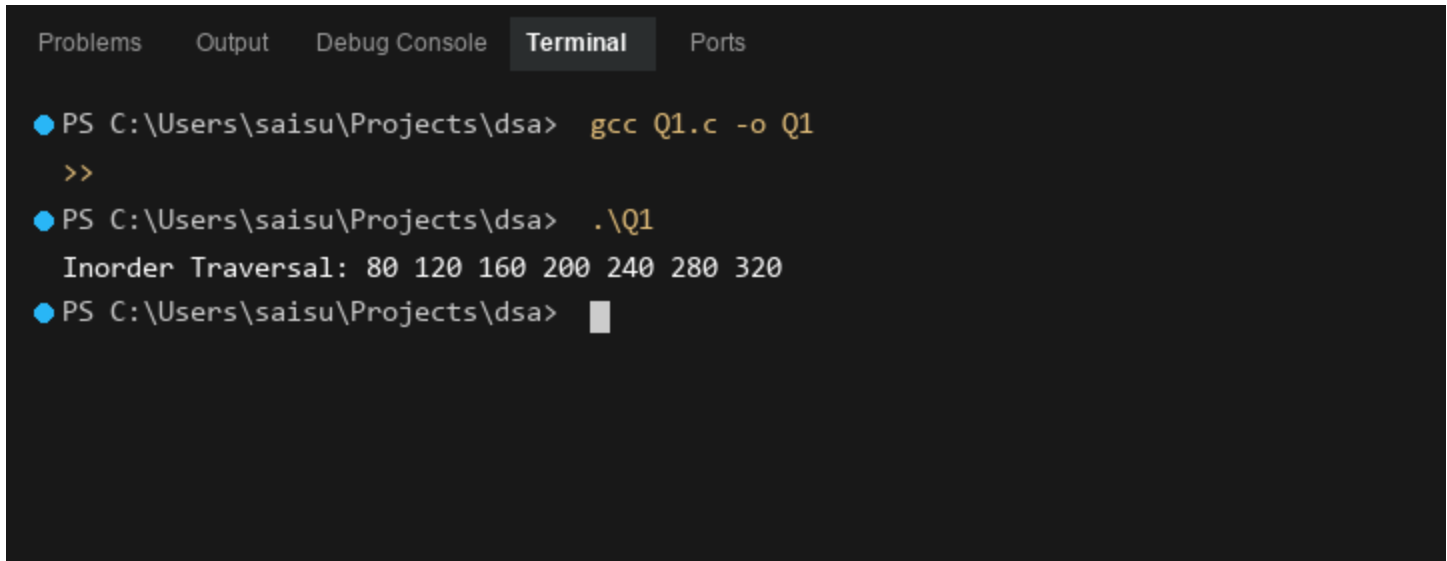
Question 1: Traversals of Binary Search Tree

Problem Description:

Construct a Binary Search Tree (BST) from given keys and implement recursive Inorder (L-Root-R), Preorder (Root-L-R), and Postorder (L-R-Root) depth-first traversals.

Sample Input: 200 120 80 160 280 240 320 | Sample Output: Inorder: 80 120 160 200 240 280 320

Output Screenshot:



```
Problems  Output  Debug Console  Terminal  Ports

PS C:\Users\saisu\Projects\dsa> gcc Q1.c -o Q1
>>
PS C:\Users\saisu\Projects\dsa> .\Q1
Inorder Traversal: 80 120 160 200 240 280 320
PS C:\Users\saisu\Projects\dsa> 
```

Question 2: Level Order Traversal of Binary Tree

Problem Description:

Perform Breadth-First Level Order Traversal on a binary tree using a FIFO queue to visit nodes level-by-level from top to bottom.

Sample Input: Root=1, Left=2, Right=3 | Sample Output: Level Order: 1 2 3

Output Screenshot:



The screenshot shows a terminal window with the following content:

```
Problems  Output  Debug Console  Terminal  Ports

● PS C:\Users\saisu\Projects\dsa> gcc Q2.c -o Q2
  >>
● PS C:\Users\saisu\Projects\dsa> .\Q2
  Level Order: 1 2 3
● PS C:\Users\saisu\Projects\dsa> █
```

Question 3: Calculate Height and Depth of a Binary Tree

Problem Description:

Find the maximum height/depth of a binary tree (number of nodes along longest root-to-leaf path) using postorder recursive calculation.

Sample Input: Binary Tree with 3 levels | Sample Output: Height of Tree: 3

Output Screenshot:



```
Problems  Output  Debug Console  Terminal  Ports

● PS C:\Users\saisu\Projects\dsa> gcc Q3.c -o Q3
>>
● PS C:\Users\saisu\Projects\dsa> .\Q3
Height of Tree: 3
● PS C:\Users\saisu\Projects\dsa> █
```

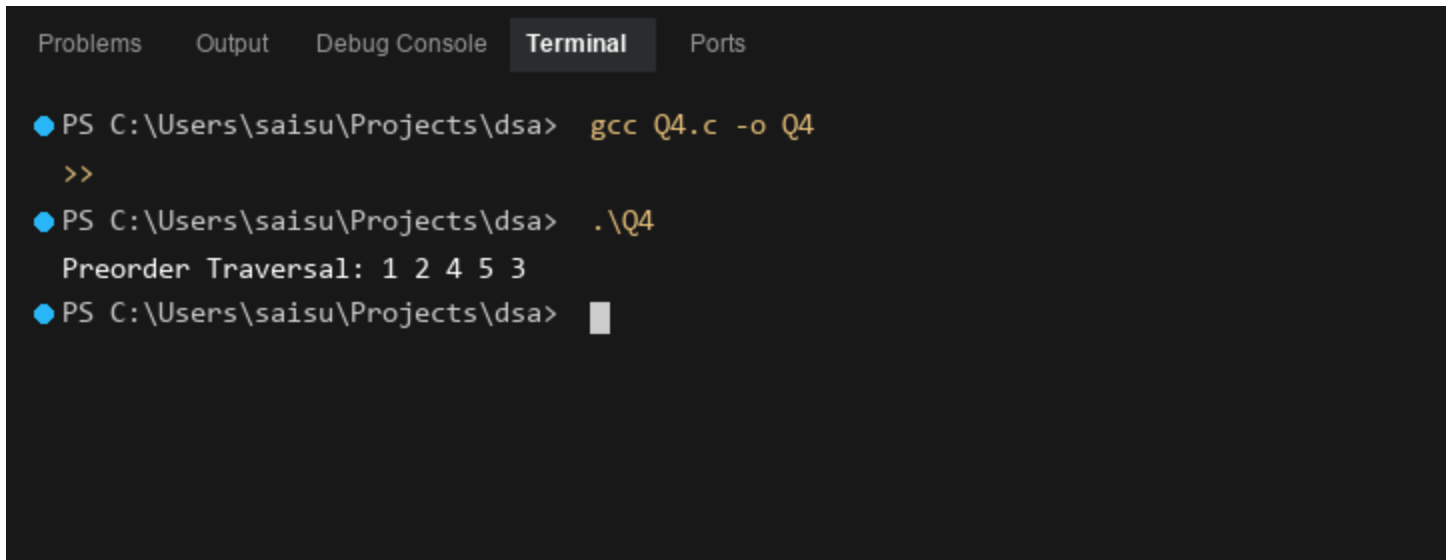
Question 4: Construct Binary Tree from Inorder and Postorder

Problem Description:

Reconstruct a unique binary tree given its Inorder and Postorder traversal sequences and display its generated Preorder traversal.

Sample Input: Inorder: 4 2 5 1 3, Postorder: 4 5 2 3 1 | Sample Output: Preorder: 1 2 4 5 3

Output Screenshot:



```
Problems  Output  Debug Console  Terminal  Ports

● PS C:\Users\saisu\Projects\dsa> gcc Q4.c -o Q4
  >>
● PS C:\Users\saisu\Projects\dsa> .\Q4
  Preorder Traversal: 1 2 4 5 3
● PS C:\Users\saisu\Projects\dsa> █
```

Question 5: Segment Tree Range Minimum Query (RMQ)

Problem Description:

Build a Segment Tree over an array to answer Range Minimum Queries (RMQ) and support logarithmic point updates in $O(\log N)$ time.

Sample Input: Array [2, 5, 1, 4, 9, 3], Query Range [1, 4] | Sample Output: Minimum in range: 1

Output Screenshot:



```
Problems  Output  Debug Console  Terminal  Ports

● PS C:\Users\saisu\Projects\dsa> gcc Q5.c -o Q5
>>
● PS C:\Users\saisu\Projects\dsa> .\Q5
Min in range [1, 4]: 1
● PS C:\Users\saisu\Projects\dsa> █
```

Question 6: Lowest Common Ancestor in Binary Search Tree

Problem Description:

Find the Lowest Common Ancestor (LCA) of two given nodes n_1 and n_2 in a Binary Search Tree by checking split key values in $O(h)$ time.

Sample Input: BST with keys 20, 8, 22, 4, 12, 10, 14; LCA(10, 14) | Sample Output: LCA: 12

Output Screenshot:



```
Problems  Output  Debug Console  Terminal  Ports

● PS C:\Users\saisu\Projects\dsa> gcc Q6.c -o Q6
>>
● PS C:\Users\saisu\Projects\dsa> .\Q6
LCA of 10 and 14: 12
● PS C:\Users\saisu\Projects\dsa> █
```

Question 7: Diameter of a Binary Tree

Problem Description:

Calculate the diameter (length of the longest path between any two leaf nodes) of a binary tree in linear $O(N)$ time.

Sample Input: Binary Tree of 5 nodes | Sample Output: Diameter: 4

Output Screenshot:



```
Problems  Output  Debug Console  Terminal  Ports

● PS C:\Users\saisu\Projects\dsa> gcc Q7.c -o Q7
>>
● PS C:\Users\saisu\Projects\dsa> .\Q7
Diameter of Tree: 4
● PS C:\Users\saisu\Projects\dsa> █
```

Question 8: Binary Indexed Tree (Fenwick Tree) Prefix Sum

Problem Description:

Implement Fenwick Tree (BIT) supporting $O(\log N)$ prefix sum queries and point updates using lowbit (i & -i) arithmetic.

Sample Input: Array: 2 1 1 3 2 3 4 5 6 7 8 9 | Sample Output: Prefix Sum up to index 5: 12

Output Screenshot:



```
Problems  Output  Debug Console  Terminal  Ports

● PS C:\Users\saisu\Projects\dsa> gcc Q8.c -o Q8
>>
● PS C:\Users\saisu\Projects\dsa> .\Q8
Prefix Sum of first 5 elements: 9
● PS C:\Users\saisu\Projects\dsa> █
```

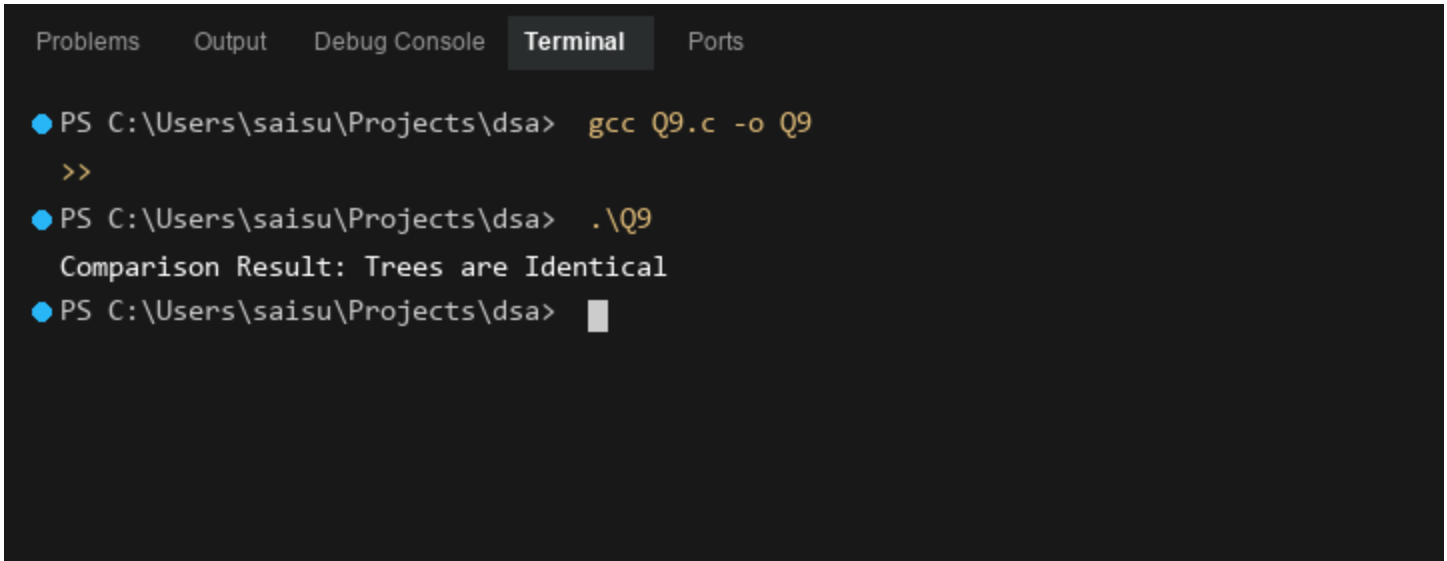
Question 9: Check if Two Binary Trees are Identical

Problem Description:

Given roots of two binary trees, write an algorithm to check whether they are structurally identical and have the same node values.

Sample Input: Tree1: [1, 2, 3], Tree2: [1, 2, 3] | Sample Output: Trees are Identical

Output Screenshot:



```
Problems  Output  Debug Console  Terminal  Ports

● PS C:\Users\saisu\Projects\dsa> gcc Q9.c -o Q9
  >>
● PS C:\Users\saisu\Projects\dsa> .\Q9
  Comparison Result: Trees are Identical
● PS C:\Users\saisu\Projects\dsa> █
```

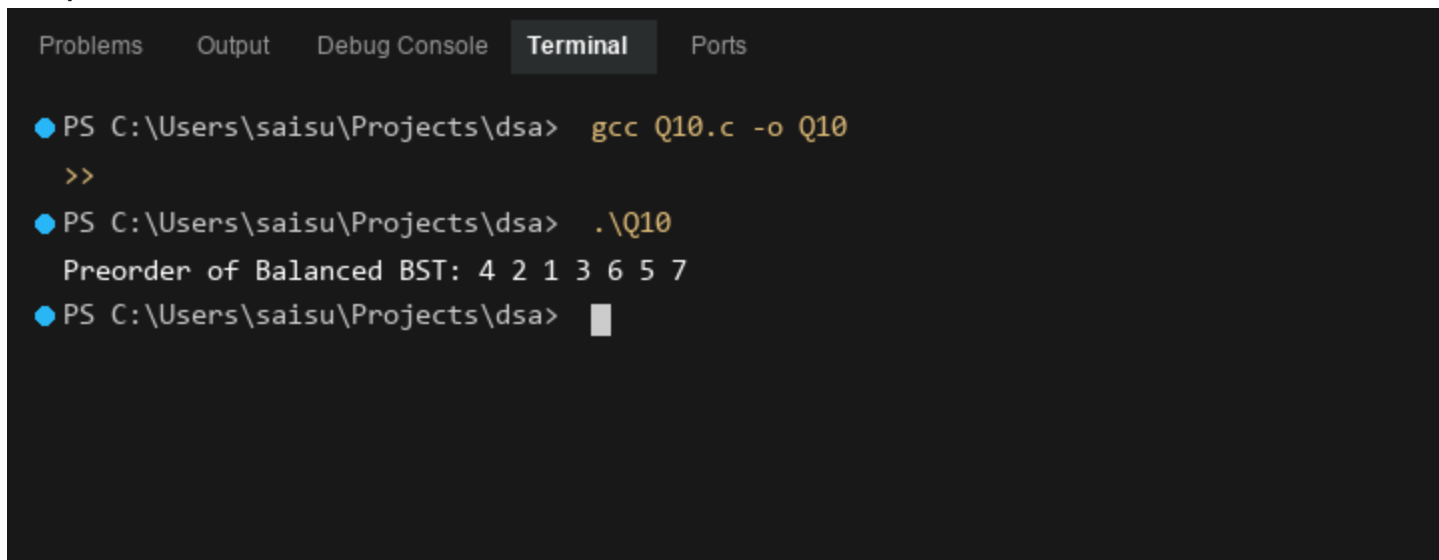
Question 10: Convert Sorted Array to Balanced BST

Problem Description:

Given a sorted array of distinct elements, convert it into a height-balanced Binary Search Tree where depth of two subtrees differs by at most 1.

Sample Input: 1 2 3 4 5 6 7 | Sample Output: Root: 4, Preorder: 4 2 1 3 6 5 7

Output Screenshot:



```
Problems  Output  Debug Console  Terminal  Ports

● PS C:\Users\saisu\Projects\dsa> gcc Q10.c -o Q10
  >>
● PS C:\Users\saisu\Projects\dsa> .\Q10
  Preorder of Balanced BST: 4 2 1 3 6 5 7
● PS C:\Users\saisu\Projects\dsa> █
```